The Climate Standoff: *A Dynamic Game Model of Strategic Tipping in Global Climate Coordination*

Abstract

This paper develops a finite-horizon dynamic Markov game of global climate coordination among four heterogeneous blocs: the United States, China, the European Union, and a Rest of World aggregate comprising large emerging economies — Brazil, India, Mexico, Vietnam, and their peers — at intermediate stages of industrial development. Each period, blocs choose whether to adopt a low-carbon transition; adoption is irreversible. Transition costs fall endogenously through learning spillovers, political pressure rises with coalition size, and collective benefits accrue once a weighted participation threshold is crossed. I characterise behaviour using quantal response equilibrium (QRE), which replaces the knife-edge solutions of Nash-based coalition models with smooth probabilistic choice, permitting well-behaved comparative statics across the full parameter space. I recover cardinal payoff weights by Simulated Method of Moments, anchoring the model to the 2025 strategic environment.

The analysis proceeds through two complementary experiments. Marginal parameter sweeps identify how far each strategic lever must move to compress the probability of coordination failure. Global sensitivity analysis then jointly perturbs the full parameter space across 1,000 draws, testing whether those lever rankings survive simultaneous variation. Both experiments are complemented by functional-form robustness checks.

Three findings are robust across specifications. First, jointly extending the effective planning horizons of the United States and China is the most powerful single intervention: it is the only lever that drives coordination success to near-certainty under a modest bilateral improvement. Second, China's domestic transition cost burden is an independent bottleneck that longer horizons alone cannot resolve; targeted cost reduction through technology transfer or concessional finance is a necessary complement. Third, perceived feasibility is itself a coordination determinant. Institutions that make partial progress visible accelerate adoption without altering underlying incentives.

A novel asymmetry emerges from the counterfactual analysis. Reducing volatility in US climate policy improves coordination outcomes both directly and by disciplining Chinese strategic expectations. Greater Chinese policy coherence does not reciprocate this effect on US expectations. The asymmetry survives parameter swaps between the two blocs, which suggests it reflects a structural property of asymmetric coordination games rather than an artefact of calibration.

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Introduction:

Threshold public goods models explain coordination failure through two interlocking mechanisms: collective benefits depend on reaching a critical participation mass, and multiple equilibria make that mass difficult to attain. The literature has produced elegant characterisations of why cooperation stalls. It has been less successful at explaining why it sometimes accelerates.

The most consequential shifts in recent climate policy were not preceded by formal treaties. The European Union's Carbon Border Adjustment Mechanism did not emerge from multilateral negotiation; it became politically viable only after domestic decarbonisation created the incentive to export climate standards through trade policy. The United States Inflation Reduction Act reshaped global clean-energy investment through large-scale subsidies, triggering competitive policy responses across Europe and Asia within eighteen months. In both cases, early movers altered the payoff matrix for everyone else, and the institutional architecture followed strategic logic rather than producing it.

The strategic environment is most usefully understood as a multilateral game of chicken among highly asymmetric players. The blocs are not driving identical cars toward the same obstacle. A small number are systemically important enough that their inaction carries disproportionate global consequences, placing them under asymmetric pressure to move first. Others face weak early-mover incentives because short political horizons and high transition costs raise the price of unilateral action. Still others sit closer to the cliff: climate damages arrive faster, hit more vulnerable sectors, and leave less room to wait. The coordination problem is therefore shaped not only by whether blocs eventually cooperate, but by who can afford to delay and whose move most strongly updates the incentives of every other player.

Standard treaty-based models abstract from precisely these features. By treating enforcement pressure, cost trajectories, and threshold progress as exogenous rather than as outcomes of strategic interaction, they cannot account for the timing, escalation, and asymmetry that define modern climate geopolitics. A framework that endogenises those variables can explain why cooperation sometimes deepens even when formal agreements stagnate.

The paper proceeds as follows. We first review the relevant literature across three strands: coalition stability and equilibrium concepts, strategic tipping and diffusion dynamics, and the gaps both leave open. The formal model is then developed, followed by the calibration strategy, which separates the ordinal structure identified from observable data from the cardinal scaling recovered by Simulated Method of Moments. The results of the marginal parameter sweeps and global sensitivity analysis are then reported, from which the policy implications are drawn. An appendix provides equilibrium uniqueness verification, functional form robustness checks, and full sensitivity tables.

Review of Literature

Equilibrium Concepts : Nash to Quantal Response

The dominant solution concepts in the climate cooperation literature share a common assumption: players best-respond exactly to the strategies of others. Barrett (1994) uses stage-game Nash equilibrium and renegotiation-proof repeated-game equilibrium to characterise self-enforcing international environmental agreements. Harstad (2016) selects Markov perfect equilibrium on the grounds that strategies depending only on payoff-relevant state variables are both analytically tractable and behaviourally plausible. The assumption of exact best response is not incidental to these frameworks. It is load-bearing.

Carraro and Siniscalco (1993) establish the classical fragility result: stable coalitions under internal and external stability conditions are typically small, and because participation decisions are discrete, small parameter changes induce large discontinuous shifts in equilibrium coalition size. The framework moves abruptly between near-full cooperation and near-complete defection, which limits its usefulness precisely where the empirical record is most interesting: the margin where marginal incentive changes produce cascading adoption rather than stasis.

Quantal response equilibrium resolves this by replacing exact best response with probabilistic choice (McKelvey and Palfrey, 1995). Players choose better options more often, but not always, with choice probabilities increasing in expected payoffs governed by the rationality parameter λ. A marginal increase in adoption costs therefore reduces adoption probability incrementally rather than flipping participation off entirely. Goeree, Holt, and Palfrey (2003) show that QRE closely tracks observed behaviour across a wide range of laboratory strategic environments.

The calibrated value of λ carries direct substantive content. Putnam (1988) formalises why governments cannot behave as frictionless unitary optimisers: international commitments must remain feasible in the domestic ratification game. In this setting, stochastic choice is a reduced-form representation of those political frictions rather than purely behavioural error. The noise in λ reflects domestic win sets, bureaucratic friction, and electoral uncertainty — not irrationality.

Practical uniqueness is supported by convergence to identical fixed points across multiple initialisations, suggesting the solution is not sensitive to starting conditions. The appendix reports full convergence diagnostics.

QRE ensures smooth changes in incentives produce smooth changes in behaviour. It does not explain when those smooth responses accumulate into self-reinforcing adoption cascades. That requires a theory of tipping.

Strategic tipping and diffusion dynamics

Quantal response equilibrium explains how equilibrium behaviour responds smoothly to shifting incentives. It does not explain when smooth responses accumulate into cascades. A separate body of work addresses this directly.

The physical and technological foundations of tipping are reasonably well understood. Way et al. (2022) show that endogenous cost declines through fast learning curves for renewables can shift mitigation pathways more sharply than most integrated assessment models assume, a finding consistent with the directed technical change literature establishing that innovation incentives respond to deployment scale in ways that can generate self-reinforcing cost trajectories (Acemoglu et al., 2012). Farmer et al. (2019) introduce sensitive intervention points in socioeconomic systems, arguing that small, well-targeted interventions can exploit these amplification mechanisms to trigger cascading low-carbon transitions. Otto et al. (2020) identify specific social subsystems where positive feedback could rapidly compress emissions trajectories, and Sharpe and Lenton (2021) extend this into a framework of upward-scaling tipping cascades in which activating one socio-technical tipping point raises the probability of triggering the next. Milkoreit et al. (2018) formalise the social tipping point concept across interdisciplinary contexts, identifying belief shifts and perceived feasibility as distinct triggers alongside cost and technology channels.

These contributions capture dynamics observed in real energy transitions. They share a structural limitation. Adoption decisions are embedded in aggregate diffusion mechanisms, systems feedback frameworks, or reduced-form cost trajectories rather than arising from explicit strategic interaction between heterogeneous countries. This matters because the mechanism is difficult to identify. Comin and Hobijn (2010) document precisely this puzzle in a long historical panel of technology diffusions: structurally similar countries adopt at systematically different speeds, with institutional and political factors explaining residual variation that cost differentials alone cannot. Without an explicit strategic foundation, it is difficult to distinguish whether cascades emerge from cost feedbacks, expectation shifts, or politically mediated peer effects — and the policy implications of each differ sharply.

A second problem compounds the first. Tipping behaviour in formally specified models is often sensitive to chosen parameter values, making it difficult to distinguish structurally robust dynamics from calibration artefacts. A finding that holds only at a particular cost trajectory or rationality parameter is not a finding about tipping. It is a finding about one point in the parameter space.

Taken together, these gaps define the space this paper occupies. The literature needs a framework that endogenises tipping through asymmetric strategic interaction, and a disciplined method for establishing which tipping dynamics survive joint parameter perturbation. The next section turns to both.

Gaps and Contribution

Two gaps run through the literature reviewed above. The first is structural. Existing frameworks rarely model declining costs, endogenous political pressure, and asymmetric influence as joint strategic determinants of cooperation. Agreement-based models treat enforcement as exogenous: Barrett (1994) characterises self-enforcing agreements under fixed participation incentives, and Nordhaus (2015) demonstrates that coalition models generate stable but small cooperative groups precisely because the strategic environment outside the coalition is held constant. Diffusion models embed adoption in aggregate cost functions or reduced-form trajectories rather than strategic interaction (Grubb et al., 2021). Neither tradition can explain how early adoption by anchor players reshapes the incentives of strategic holdouts over time, because neither gives those players an explicit strategic role.

The second gap is methodological. Game-theoretic models of climate cooperation are known to be sensitive to parameterisation. Kolstad (2007) documents that quantitative results in the IEA literature shift substantially across plausible cost and benefit assumptions, yet few studies distinguish the cross-bloc structure pinned down by observable data from the latent cardinal strength of the underlying payoff channels. Most computational approaches calibrate to a single benchmark or rely on ad hoc sensitivity checks, making it difficult to tell whether a finding reflects a structural property of the game or a feature of one point in the parameter space.

This paper addresses both gaps by embedding these mechanisms in a finite-horizon dynamic Markov game with heterogeneous blocs and irreversible adoption. Behaviour is characterised using quantal response equilibrium, which allows smooth comparative statics and realistic strategic delay without the brittleness of Nash-based coalition models.

The methodological contribution lies in separating calibration into two layers. The ordinal layer is identified directly from macroeconomic and emissions data, preserving the empirical asymmetries across blocs without imposing arbitrary structure. The cardinal layer is then recovered using simulated method of moments (McFadden, 1989), matching observable features of the contemporary climate transition: EU leadership, strategic free-riding by major emerging economies, and the delayed onset of self-reinforcing adoption. Together these produce a single fitted strategic baseline anchored to the 2025 strategic environment.

From that baseline, the analysis identifies which policy levers most reliably advance earlier coordination. Local sweeps establish which channels matter in isolation. Global sensitivity analysis (Saltelli et al., 2008) then stress-tests the rankings by jointly perturbing the full parameter space, separating levers that are locally powerful from those that remain influential once the wider strategic environment moves. A lever that works only near the baseline calibration is not a robust prescription.

Model Setup / Math (see overleaf doc)

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Calibration

Implementing the model requires assigning numerical values to all parameters. Some can be disciplined directly by observable data. Others cannot. The calibration strategy addresses this by separating the problem into two layers, each with a distinct identification logic.

The first layer fixes ordinal structure: the relative positioning of blocs within each parameter channel. Which bloc faces the highest transition costs, which is most exposed to climate damages, which exerts the greatest systemic influence over others — these cross-bloc rankings can be inferred from data. World Bank (2026) development indicators, national emissions inventories, and bilateral trade statistics provide consistent evidence on the relative positions of the United States, European Union, China, and the Rest of the World. This layer is treated as empirically identified and held fixed throughout the analysis. Perturbing it would mean contradicting the observable record.

The second layer recovers cardinal scaling: the absolute magnitude of each payoff channel and its weight in shaping strategic behaviour. The relative importance governments place on transition costs versus climate damages versus political pressure is not directly observable and cannot be read off macroeconomic data (Nordhaus, 2007). Rather than treating these weights as free scenario assumptions, I estimate them using simulated method of moments (McFadden, 1989).

The moments are chosen to represent the observable strategic fingerprint of the 2025 climate environment. I target the renewable adoption ratios between the EU and its peers, the near-certain probability of EU first-mover behaviour, and the absence of near-term global coordination cascades. Each moment captures a distinct feature of the strategic landscape: the cost channel leaves a fingerprint in adoption gaps, leadership probabilities identify the pressure channel, and the absence of cascades pins down the threshold structure. Together they anchor the model to the contemporary environment rather than to a stylised benchmark.

The fitted parameter vector is therefore the one that makes the model reproduce these observable features of the world. Ordinal structure is data-identified. Cardinal scaling is structurally estimated. The sensitivity analysis in the results section is conducted around this fitted baseline, which is what allows the distinction between locally powerful levers and structurally robust ones to carry empirical weight.

Layer 1 : Ordinal Structure

Per Player Params:

Each per-player parameter is constructed from bloc-level data through a structural transformation anchored to the EU as a reference bloc. This preserves empirical ordinal relationships across blocs while a single scaling parameter alpha determines the absolute level.

Transaction Costs:

Transition costs are derived from the carbon intensity of GDP, constructed from World Bank (2026) emissions and output data, adjusted for development. Carbon intensity captures the extent of structural change required for decarbonisation: higher-intensity economies must replace more emissions-intensive capital to meet the same mitigation target. Raw intensity cannot be taken at face value for less developed economies, however. Low carbon intensity in those cases often reflects limited industrialisation rather than genuinely clean infrastructure, which would falsely register as low transition costs.

To correct for this, I apply a development-gated interpolation. Each bloc's effective carbon intensity is defined as a weighted average of its observed intensity and a global baseline, where the weight rises with GDP per capita relative to the richest bloc, with all underlying series drawn from World Bank (2026) national accounts and development indicators. Developed economies retain their observed advantage. Less developed economies are partially adjusted toward the baseline, preventing the model from treating structural underdevelopment as a decarbonisation head start.

An affordability penalty is then applied on top. The cost of capital for renewable energy projects in developing economies is empirically two to three times higher than in advanced economies, reflecting financing constraints and weaker deployment capacity (Ameli et al., 2021). The affordability elasticity (epsilon $\epsilon = 0.3$) governs how steeply effective costs rise as income falls below the richest bloc. Together the two adjustments ensure the ordinal cost structure reflects genuine transition burdens rather than artefacts of the raw intensity data.

Climate damages: Climate damages are proxied by the share of agriculture, forestry, and fishing in GDP, drawn from World Bank (2026) national accounts data. This captures structural economic exposure to climate risk: economies with larger primary sectors face greater direct losses from temperature extremes, water stress, and ecosystem disruption, consistent with Burke et al. (2015) who document that agricultural productivity losses account for the largest share of measurable climate damages across countries at varying income levels. More composite vulnerability indices exist — the ND-GAIN Country Index being the most widely used — but agriculture share is preferred for its direct observability and consistency across blocs. The proxy is a deliberate choice between precision and comparability, and comparability wins at this level of aggregation.

Political pressure: Political pressure is proxied by trade openness, measured as trade as a share of GDP. A more trade-dependent economy has more to lose from non-cooperation: carbon border adjustments,

climate-linked conditionality, and broader diplomatic friction all arrive through trade relationships. Trade openness therefore captures the degree to which a bloc's economic welfare is hostage to the policy choices of its major partners, and how exposed it is to externally imposed costs if it delays.

Influence weights: Influence weights are defined as a GDP-weighted combination of emissions share and GDP share, with weights of 0.30 and 0.70 respectively, constructed entirely from World Bank (2026) national accounts and emissions data. Emissions share captures a bloc's necessity for any effective global agreement — a coalition excluding major emitters cannot stabilise the climate regardless of its economic composition. GDP share captures the economic mechanisms through which adoption influence operates: a wealthier adopter drives larger cost reductions through supply chain scale and learning spillovers, exerts stronger trade-linked pressure on holdouts, and shifts the threshold calculus more substantially through its economic footprint. The 70/30 weighting reflects the judgement that influence over coordination dynamics flows primarily through economic channels rather than through emissions necessity alone. This weighting is varied in the robustness checks reported in the appendix.

All parameters are normalised relative to the EU benchmark, so each scaling parameter alpha Directly sets the EU's level for a given channel. The values for other blocs are then determined by their data-implied ratios to the EU. This structure limits the modeller's degrees of freedom to the choice of alpha, while the cross-bloc relationships remain fully data-driven. The calibration problem reduces to recovering the vector of alpha values that make the model reproduce the target moments which is precisely what the SMM procedure does.

Special Note on Player Discount Rates

Discount factors follow Nordhaus (2007), reflecting observed political and economic time horizons rather than prescriptive ethics. A game-theoretic framework concerned with strategic behaviour belongs on the descriptive side of this debate. Each bloc's discount factor combines pure time preference with institutional commitment credibility, and because each model period spans five years, delta represents a compounded annual rate rather than a per-period one.

The EU receives $\delta = 0.85$ (3.25% annually), reflecting the legal bindingness of the European Climate Law and over two decades of continuous ETS operation. China is assigned $\delta = 0.80$ (4.6%), where centralised governance supports continuity through the five-year plan system but growth constraints shorten the effective horizon. The US receives $\delta = 0.75$ (5.9%), incorporating a reversal premium for the institutional instability illustrated by withdrawal from and re-entry into the Paris Agreement. The Rest of the World is assigned δ\delta $\delta = 0.70$ (7.7% annually), reflecting the sovereign borrowing costs of major developing economies including India, Indonesia, Brazil, and Mexico. World Bank (2026) international debt statistics report nominal sovereign yields for these economies ranging from approximately 6% to 10% annually, with a GDP-weighted average around 7.5%. This is higher than advanced economy rates but meaningfully lower than the least developed economies, consistent with the bloc's redefinition as middle-income developing economies facing genuine but not structurally impossible adoption tradeoffs (Ameli et al., 2021).

The ordering EU > China > US > RoW holds across all specifications and generates the central strategic tension the model is designed to illuminate: the blocs with greatest patience are not those with greatest climate vulnerability.

Global Params

Global parameters govern the structure of the game rather than the characteristics of individual players. They cannot be identified through cross-bloc comparisons, so each is anchored to a baseline drawn from published estimates in the climate economics and game theory literature. These baselines are reference points, not fixed values. The global parameters are treated as experimental inputs and varied systematically across the analysis.

Three structural features of the game turn on these parameters. The threshold and curvature of the collective benefit function determine whether coordination emerges gradually or through sharp tipping. The learning rate governs how quickly transition costs decline as adoption spreads. The discount rate captures how strongly time preferences shape the tradeoff between early adjustment costs and future stabilisation benefits. Together they define the strategic terrain on which the per-player asymmetries play out. The sensitivity analysis in the results section stress-tests the findings across the range of plausible values for each.

Coordination Threshold

The coordination threshold (theta = 0.80) defines the weighted participation level at which stabilisation benefits become substantial. The Paris Agreement's 55% emissions threshold governs entry into force, but entry into force is not effective mitigation. Net-zero pathways in IEA (2021) and IPCC 1.5C scenarios consistently require near-universal engagement among major emitters. The four blocs modelled here account for the substantial majority of global emissions, so a threshold of 0.80 reflects the near-universal requirement those pathways imply. As with other global parameters, tau is varied systematically in the sensitivity analysis to examine how the stringency of the coordination requirement shapes equilibrium outcomes.

Sigmoid Steepness

The sigmoid steepness parameter (η) determines how sharply stabilisation benefits concentrate around the threshold. High values produce tipping-point dynamics, where benefits activate rapidly once θ is crossed. Lower values generate a more gradual accumulation of benefits across adoption levels. The baseline (η = 15) implies a steep transition, with S(W) rising from approximately 0.06 to 0.94 within a ±0.2 band around θ. Since this parameter governs the qualitative shape of payoffs rather than a directly observable quantity, it is treated as an experimental lever.

QRE Rationality

The precision parameter lambda governs how sharply each bloc's strategy responds to payoff differences. Higher values push behaviour toward deterministic best-response; lower values imply greater noise in strategic choice. Experimental estimates typically range from 0.5 to 10 depending on context (McKelvey and Palfrey, 1995; Goeree, Holt, and Palfrey, 2003).

The baseline sets lambda = 1.5 as a common global benchmark, placing it toward the lower end of the experimental range. This reflects the persistent noise in international climate negotiations: domestic political constraints, incomplete information, electoral volatility, and bureaucratic friction all prevent governments from acting as perfectly rational best-responders even when the strategic logic of cooperation is clear.

The model also allows lambda to vary at the bloc level. Bloc-specific precision captures heterogeneity in institutional stability, political coherence, and decision-making capacity that a single global parameter cannot. This makes it possible to examine how differences in political rationality across actors shape the timing and likelihood of global coordination, rather than assuming all blocs navigate strategic uncertainty with equal precision.

Technology Learning Rate

The learning rate ($\gamma = 0.25$) captures cost reductions driven by cumulative adoption through spillovers, supply chain development, and learning-by-doing. Empirical estimates suggest cost declines of 20–25% per doubling of capacity for solar PV and 10–15% for wind (Rubin et al., 2015; IRENA, 2023). The chosen value aligns with the upper end of these estimates, reflecting that "adoption" in the model represents a broad technological transition rather than a single technology.

Damage Escalation

The damage escalation rate ($\kappa = 0.05$) governs how climate damages compounds under delayed action. Over the model's 50-year horizon, this implies an increase of roughly 50% in baseline damages under persistent non-cooperation. This magnitude is consistent with IPCC AR6 estimates suggesting losses on the order of 1–2% of GDP per decade of delayed mitigation.

Layer 2 : Simulated Method of Moments (SMM) Calibration

The cardinal scaling layer is estimated using a simulated method of moments. Four scaling parameters remain free once the ordinal structure is fixed: transition costs, climate damages, political pressure, and coordination benefits. SMM recovers their relative magnitudes by selecting the parameter vector that minimises the weighted squared distance between five model-implied moments and their empirical counterparts.

All other structural parameters are held fixed during estimation, including bloc-specific discount factors, rationality parameters, sigmoid steepness, and damage escalation. This isolates identification onto the four scaling parameters and prevents the cardinal weights from absorbing variation driven by unrelated assumptions. For any candidate vector, the game is solved under quantal response equilibrium using backward induction, and the resulting strategies are passed through Monte Carlo simulation to generate model-implied moments.

SMM Targets vs Model

[TABLE DISPLAYING Fitted Baseline vs Targets]

Targeted Moments

Six moments are targeted, chosen because each provides distinct identifying variation for the underlying payoff channels.

- Moment 1 targets the EU-US adoption gap in period one at 0.25. The EU has been the clear leader in clean energy transition over the past two decades. The US has not matched this pace despite comparable economic scale, with clean energy policy advancing and reversing across administrations rather than compounding (Wurzel et al., 2019; Falkner, 2016).

- Moment 2 targets US period-one adoption probability at 0.10. The US has not sustained a consistent national clean energy transition. Federal climate policy has shifted dramatically across administrations, and renewable capacity deployment remains well behind the EU on a comparable economic basis, as World Bank (2026) energy transition indicators confirm.

- Moment 3 targets the EU lead ratio at 2.5x. The EU has moved substantially faster than any peer bloc on clean energy transition. Renewable energy share, carbon pricing coverage, and the pace of industrial decarbonisation all place the EU clearly ahead of the US, China, and other major economies by observable measures (Backstrand and Elgstrom, 2013; IRENA, 2023).

- Moment 4 targets China's period-one adoption probability at 0.05. China remains the world's largest emitter and largest coal consumer. Despite rapid renewable buildout, fossil fuel expansion has continued alongside it and no binding near-term transition commitment has been made (IEA, 2025; WRI, 2025).

- Moment 5 targets the probability of zero adoption by period two at 0.30. Despite the EU's sustained climate leadership, no other major economy has made a credible near-term transition commitment. The US, China, and major developing economies all face significant barriers to early adoption, making a world where the second policy cycle begins without any additional commitment a realistic baseline rather than a tail event.

- Moment 6 targets mean coordination timing conditional on success at 5.0 periods, corresponding to roughly 2045 to 2050. Net-zero pathways in IEA (2021) and IPCC 1.5°C scenarios consistently identify mid-century as the realistic window for meaningful global coordination among major emitters. Targeting this moment disciplines the model against implausibly fast coordination while remaining consistent with the international frameworks the blocs have nominally endorsed.

Identification of Params, methodology and limitations

Six moments identify four unknown cardinal weights, rendering the system over-identified by two restrictions. Under exact identification the optimiser always finds a perfect fit regardless of whether the parameters are genuinely identified. The additional moments force the estimator to find a single parameter vector fitting all six simultaneously, which is only possible if the underlying channels are doing real work.

Minimisation uses the Nelder-Mead algorithm (Nelder and Mead, 1965). Gradient-based methods are unsuitable because the objective is evaluated through Monte Carlo simulation, producing a noisy objective surface that violates differentiability assumptions. Nelder-Mead evaluates the objective directly without requiring gradient information. To guard against local minima the procedure is run from multiple random starting points with the global minimum taken as the estimate.

The four parameters are not equally precisely identified. alpha_c αc and alpha_d αd are the most tightly identified. alpha_b αb is identified within a broader region. alpha_p αp recovers at its lower bound, indicating the moment set does not provide sufficient variation to pin down the pressure channel cardinal. This is consistent with the economic logic of the adoption decision: transition costs represent direct and immediate structural burdens while political pressure operates through more diffuse channels, making it unlikely that period-one adoption snapshots contain strong identifying variation for pressure scaling. The policy hierarchy reported in the results section is robust to this limitation — forcing alpha_p αp to match alpha_c αc in cardinal scale leaves the ranking of patience, cost, and threshold channels unchanged. Full details are reported in the appendix. The estimates characterise a convergence region rather than a unique point and are interpreted accordingly throughout the analysis.

[PARAM VECTOR] TOO + COMMENTARY

Verification and Stability

Five verification checks are applied following estimation.

The first check restarts the optimiser from the estimated solution. Parameters shift by a maximum of 0.000026 with negligible objective improvement, confirming a genuine local minimum.

The second check reruns the model across five random seeds. Coordination success rates are stable with standard deviation 0.013, confirming Monte Carlo stability.

The third check verifies that estimated parameters preserve all five empirically-grounded ordinal rankings across blocs. All are preserved exactly.

The fourth check perturbs all six target moments by ±5% and records parameter movement. Mean elasticity of 1.52 is within the acceptable threshold of 2.0, indicating the estimates are stable to small changes in targets.

The fifth check computes the Jacobian condition number as a formal measure of identification strength, following the GMM literature (Hansen, 1982). A condition number below 100 indicates the moment vector provides genuinely distinct identifying variation rather than near-collinear signals (Belsley, Kuh, and Welsch, 1980). The six-moment specification produces a condition number of 53.6, comfortably within that bound.

[Verification and stability test table]

Experimental Design

What this model is for

The model is not designed as a forecasting tool. It does not attempt to predict whether real-world climate coordination will succeed or fail, and its outputs should not be interpreted as literal probabilities of observed outcomes. Instead, its purpose is mechanism exploration. Starting from the SMM-fitted strategic environment, the objective is to identify which policy channels exert the greatest leverage over coordination dynamics and whether those rankings remain stable once multiple structural assumptions are perturbed.

The experimental design follows directly from this objective. Each experiment begins from the fitted baseline economy, perturbs one or more structural parameters, and records how coordination success, timing, and bloc-level behavior shift in response. The shape of these responses reveals the relative sensitivity of the system to different strategic levers and provides a policy map of which channels can materially reduce coordination risk.

Maringal Parameter Sweeps

The primary analytical tool is the continuous marginal parameter sweep. One parameter is varied continuously across a predefined range while all remaining parameters are held at their SMM-estimated baseline values. At each point, the full dynamic game is re-solved and Monte Carlo simulation records the coordination success rate. The result is a marginal success curve with the parameter value on the horizontal axis and the probability of successful coordination on the vertical axis.

These curves answer a fundamental policy question: how far a given strategic lever must move before the system changes regime. A steep curve implies that small deviations from the fitted baseline generate large changes in coordination probability. A flat curve implies that the channel has limited leverage even under substantial perturbation.

Two summary statistics characterise each curve. The 90% crossing is the percentage deviation from the SMM-fitted baseline required to raise coordination success above 90%, and the 95% crossing is the deviation required to make coordination failure very unlikely. Percentage deviations are computed relative to the SMM-fitted baseline value for each parameter, so a crossing of +10% means the parameter must increase by 10% of its baseline value to reach that frontier. Where parameters have natural bounds, sweep ranges are set to respect them: discount factors are capped at 0.95 to preserve discounting, and cost multipliers are bounded below at zero. Crossings that would require exceeding these bounds are recorded as absent rather than extrapolated.

The channels swept include both bloc-specific and global structural levers: discount factors individually and jointly for the United States and China, cost multipliers, pressure multipliers, the technology learning rate, and the coordination threshold. Together these sweeps provide a detailed elasticity map of the fitted climate coordination economy and identify which strategic levers are capable of shifting the system toward safer equilibrium regions.

Global Sensitivity Analysis

Marginal sweeps isolate one channel at a time. While this cleanly identifies local leverage, it cannot establish whether these rankings remain robust once multiple structural forces move simultaneously. The global sensitivity analysis addresses this by jointly perturbing all major structural parameters around the SMM-fitted baseline (Saltelli et al., 2008).

One thousand parameter vectors are drawn from independent uniform distributions spanning plausible ranges for the key structural inputs. The four cardinal scaling parameters estimated by SMM are held fixed at their fitted values throughout: the GSA is designed to stress-test the structural and behavioural parameters of the game, not to re-estimate the payoff channel weights. All other major inputs are varied, including the four bloc-specific discount factors, bloc-specific cost and pressure multipliers, the technology learning rate, sigmoid steepness, QRE rationality parameters, and damage escalation. Costs and pressure are varied as multiplicative factors applied on top of their SMM-calibrated ordinal values, so the cross-bloc rankings identified in Layer 1 are preserved while the absolute magnitudes shift. Discount factor ranges are set to span empirically plausible variation: delta_{US} δUS (0.65, 0.88), delta_{EU} δEU (0.75, 0.90), delta_{CN} δCN (0.68, 0.88), and delta_{RoW} δRoW (0.55, 0.78). Full parameter ranges for all 26 inputs are reported in the appendix.

For each draw, the full dynamic game is resolved under quantal response equilibrium and 200 Monte Carlo simulations record coordination outcomes. Spearman rank correlations between each input and each outcome provide a non-parametric measure of global influence that is robust to nonlinearity and does not assume a particular functional relationship between parameters and outcomes (Iman and Conover, 1987). A parameter that remains highly ranked in both marginal sweeps and the GSA can be interpreted as a structurally robust policy lever. One that appears powerful in isolation but weak under joint variation suggests strong interaction effects or local nonlinearity around the fitted baseline.

Results

Model Baselines

Before examining parameter perturbations, the SMM-fitted baseline economy is first reported. Under the fitted strategic environment, coordination succeeds in 66.1% of simulated paths, with a mean coordination timing of 3.95 periods and a mean final coalition weight of 0.70.

The fitted world is best understood through the lens of a chicken game between asymmetric players. Rational actors usually swerve, and the model reflects this: coordination is a likely but uncertain outcome. The EU moves first at 31.1%, the US follows at 9.5%, China sits at 7.5%, and RoW is near-inactive at 0.1%. Adoption probabilities decline sharply after period four as the window for successful coordination narrows.

The EU's period-one probability remains highest, consistent with two decades of continuous climate legislation, a legally binding net-zero commitment, and the Carbon Border Adjustment Mechanism. China's non-trivial 7.5% reflects the redefined RoW bloc — with major developing economies now modelled as genuine strategic participants, China's period-one activity is more visible against a backdrop where RoW is no longer a structural zero. The strategic question is not what the EU does but whether its movement is sufficient to pull the holdouts in.

The interesting question is therefore not whether coordination happens. It is what determines the residual 33.9% tail risk of coordination failure. In a chicken game the rare outcome where neither player swerves is not just a bad outcome, it is the irreversible one. The purpose of the marginal sweeps and global sensitivity analysis is to identify which strategic levers compress that tail, separating instruments that nudge an already-likely outcome from those that push failure into a genuinely remote event.

[FIGURE: fig_sigma_over_time]

[FIGURE: fig_baseline_coalition_trajectory]

[BASELINE TABLE ADD IN LATEX]

Marginal Sweep Findings

The hierarchy is sharp. A small number of channels can actually compress the dangerous tail of coordination failure to safe levels. Most cannot.

The joint US-China patience channel is the single most powerful lever in the model. Improving both blocs' planning horizons by just 6.75% simultaneously jumps the system across both the 90% and 95% coordination frontiers at the same point. No other channel in the analysis produces this property.

The coordination threshold θ is the strongest single lever, needing only a 22.16% reduction to cross both the 90% and 95% frontiers simultaneously.

Individual patience channels reach the 90% frontier: δ_{US} needs +10.31% for 90% and +15.76% for 95%, δ_{CN} needs +13.64% for 90% but never reaches 95% alone. δ_{RoW} and δ_{EU} are absent from both frontiers.

Transition costs form a clear second tier. Global cost reduction crosses both frontiers simultaneously at -15.45%. Bloc-specific reductions require larger movements: US costs need -26.36% for 90% and -37.27% for 95%, CN costs need -70.00% for 90% but never reach 95%, RoW costs need -61.82% for 90% but never reach 95%.

Everything else falls well short. Pressure channels are absent from both frontiers. γ needs +132.72% for 90% and never reaches 95%. κ needs +92.80% for 90% and +161.80% for 95%. η is absent from both frontiers. λ_{US} needs +156.79% for 90% and +381.11% for 95%.

Figures of Key Sweeps:

[ADD IN LATEX]

Tables of Key Sweep params:

[Sweep Table]

GSA Findings

Where the marginal sweeps identify which levers compress the tail in isolation, the GSA tests whether these rankings survive once the entire strategic environment moves simultaneously. They do, but with important additions the sweeps alone could not reveal.

The coalition threshold θ and δ_US are essentially co-dominant. θ is the strongest correlation of coordination failure (rho = -0.382, $p < 0.001$) and delay (rho = +0.468, $p < 0.001$). δ_US is the strongest positive correlate of success (rho = +0.363, $p < 0.001$) and the strongest accelerator of timing (rho = -0.337, $p < 0.001$). No other inputs come close across both dimensions simultaneously.

The damage escalation parameter κ emerges as a first-order finding (rho = +0.342, $p < 0.001$). The sweeps require extreme deviations for κ to cross either frontier, confirming it operates through interaction effects rather than direct tail-risk compression, amplifying patience and cost channels once the strategic environment is already shifting.

The patience axis survives as a first-order driver. δ_RoW follows at rho = +0.145 and δ_CN at rho = +0.080, both materially accelerating timing. Cost_US is the strongest negative correlate of success after θ (rho = -0.360), with Cost_RoW at -0.181 and Cost_CN at -0.163. Cost_CN exhibits the largest negative correlation with China's first-mover probability of any parameter (rho = -0.476, $p < 0.001$), confirming transition costs as an independent bottleneck separate from the patience channel.

λ_US strongly accelerates coordination timing (rho = -0.261). λ_CN also accelerates timing, but with a substantially smaller effect (rho = -0.123). The asymmetry runs through first-mover probability: λ_CN is negatively correlated with China's own first-mover probability (rho = -0.318), meaning a more strategically precise China calculates more effectively that waiting dominates moving first, even though overall coordination still arrives slightly earlier.

η, which governs how sharply collective benefits concentrate around the participation threshold, also accelerates timing (rho = -0.193) and increases success (rho = +0.229). The more threshold-like the perceived benefits of coordination – the sharper the payoff concentration as the coalition approaches critical mass – the faster and more reliably the system tips toward cooperation. This confirms that threshold geometry is not merely a modelling convenience but a structural determinant of tipping dynamics

Pressure channels are indistinguishable from zero across all outcome dimensions. δ_EU remains a marginal driver, confirming the EU moves first because of its ordinal position rather than its discount factor.

The precise numerical thresholds reported here are specific to the fitted baseline calibration. The robust policy implication is the ordinal hierarchy, not the exact percentages

[GSA FIGURES [GSA TABLE]

Policy Implications

The model's central message is that climate coordination failure is a problem of strategic structure, not a shortage of green technology, insufficient carbon pricing, or inadequate diplomatic pressure. The game has a solution — rational players usually swerve — but the solution is fragile, and its fragility is concentrated along specific axes that standard policy frameworks systematically underweight.

The first axis is bilateral credibility between the US and China. The joint patience channel is the only lever that compresses the tail to safe levels in a single movement, and the GSA confirms US patience as the strongest individual driver of coordination success across the entire parameter space. What this captures is not simply time preference but whether the two pivotal players assign sufficient weight to future payoffs to make early adoption worthwhile. A US with a short effective planning horizon and a China that observes that constraint face a fundamentally different strategic calculus than two players whose institutions credibly extend the decision horizon beyond the immediate electoral or fiscal cycle. The rationality asymmetry sharpens this further: a more institutionally coherent US disciplines Chinese strategic expectations in ways that Chinese institutional coherence does not reciprocate. Reforms that extend and stabilise the US policy horizon are therefore doing double duty.

The second axis is China's transition cost burden, which the model identifies as an independent bottleneck entirely separate from the patience channel. The GSA confirms these are distinct constraints. Extending China's planning horizon without reducing its transition costs leaves the cost bottleneck intact. Technology transfer agreements, concessional green finance, and supply chain cooperation are not development assistance in this framing. They are direct interventions on the parameter that most strongly predicts whether China moves at all. A bilateral agreement that addresses planning horizons without addressing transition costs is solving half the problem.

The third axis is the perceived attainability of coordination itself. θ dominates every analytical test in this paper, and what it captures is not a physical participation threshold but a strategic belief: whether blocs assign meaningful probability to coordination success given the current coalition state. In a threshold game this belief is endogenous. Early progress raises perceived attainability, which accelerates further adoption, which validates the belief. Institutional architectures that make partial progress legible through staged milestones, early coalition wins, and credible roadmaps are shifting the strategic environment even without changing any underlying physics. The Paris ambition mechanism, IEA roadmaps, and sectoral coalitions are all attempts to move this belief. The model suggests they are the right instruments applied to the right variable.

Everything else operates as an amplifier rather than a primary lever. Damage urgency mechanisms, carbon border adjustments, pressure instruments, and learning spillovers all strengthen dynamics that are already moving but cannot initiate them. The model's clearest description of what current climate policy is doing wrong is deploying these instruments before the credibility and attainability conditions are in place. Sequencing matters: establish the strategic foundation first, then use amplifiers to accelerate and lock in the cascade once it begins.

Limitations

Three scope conditions bound the analysis. Each points toward a natural extension rather than threatening the core findings.

RoW Aggregation. The Rest of World bloc pools economies whose strategic incentives diverge substantially in practice. India faces a different cost schedule and domestic political economy than Brazil; ASEAN economies carry different trade exposure profiles than Mexico or Indonesia. The aggregation is defensible at this level of analysis, since the model is designed to identify structural levers rather than country-specific prescriptions and the core patience and threshold findings depend on US and Chinese behaviour rather than RoW composition. A disaggregated RoW would likely reveal that the cost bottleneck finding applies unevenly across the bloc, with some economies closer to the adoption frontier than the aggregate implies. This limitation is partially addressed by the GSA, which varies RoW cost and discount parameters across wide ranges and confirms that the lever hierarchy is not sensitive to RoW parameterisation. Full disaggregation remains a natural extension.

Irreversibility. The model treats adoption as fully irreversible, which disciplines the strategic problem cleanly but overstates the commitment value of early movers in practice. The US Paris withdrawal illustrates that nominal reversals are politically feasible even if economically costly. A richer specification would allow probabilistic reversal at a cost, introducing a credibility discount on early adoption that would weaken the patience channel findings at the margin. The asymmetry result, that US institutional coherence disciplines Chinese expectations in ways that Chinese coherence does not reciprocate, probably survives since it is driven by the structural weight of US decisions rather than their permanence. The quantitative crossing thresholds reported in the sweep analysis would however shift under partial reversibility, and the direction of that shift depends on how reversal costs are parameterised. This is a scope condition on the quantitative results rather than the qualitative policy hierarchy.

Discrete but Non-Binary Commitment. The model restricts each bloc to a binary adoption decision in each period. Real transitions proceed through intermediate stages: partial NDC commitments, sectoral decarbonisation, and graduated policy tightening all represent meaningful strategic moves that binary adoption cannot distinguish from inaction. A richer specification with discrete commitment levels, say 25%, 50%, 75%, and full adoption, would preserve the discrete strategic structure the solution method requires while allowing partial compliance strategies that characterise how China and the US have navigated climate commitments in practice. The cost bottleneck finding for China may be somewhat overstated under binary adoption: if partial adoption is available at lower cost, the threshold for Chinese movement is less demanding than the model implies. Extending the state space to multiple commitment levels is computationally costly with four blocs across multiple periods, which is the tractability reason the binary structure was adopted. Whether graduation dynamics materially alter the lever hierarchy is an open empirical question this paper cannot resolve.

Appendix

This appendix provides technical details and robustness checks that support the main analysis but would interrupt the flow of the results section if presented inline.

The first section presents equilibrium uniqueness verification, including the full multi-initialisation convergence table and spectral radius distribution across all game tree nodes. The second section examines robustness to functional form assumptions, testing alternative specifications for collective benefits, political pressure, and the cost-learning curve. The third section reports the cardinal scaling robustness test for the pressure channel, addressing whether the baseline pressure irrelevance finding reflects genuine strategic structure or a consequence of the low α_p recovered by SMM. The fourth section provides full GSA correlation tables and sweep crossing values for all parameters across all outcome dimensions.

Results in the main text are robust to the perturbations examined here. Where robustness is partial rather than complete, this is noted explicitly and the relevant tables are cross-referenced from the main text.

Equilibrium Uniqueness

Quantal response equilibrium is defined as a fixed point in the mapping from beliefs to logit best-response probabilities. Fixed-point problems of this kind can in principle admit multiple solutions, meaning the model could characterise different equilibria depending on where the iteration is initialised. If multiple fixed points exist, the findings would depend on equilibrium selection rather than the underlying strategic structure, which would undermine the policy conclusions. Two tests are applied to rule this out.

The first initialises the fixed-point iteration from 26 starting points at each of the 110 nodes with two or more active players, including the corners of the probability simplex and random interior points. All initialisations converge to identical equilibrium probability vectors, with a maximum deviation of 1.63×10^{-8} across all nodes and starting points. The second test computes the spectral radius of the Jacobian of the logit best-response map at the converged solution. A spectral radius below one confirms the fixed point is a strict contraction, meaning the iteration is guaranteed to converge to a unique solution regardless of initialisation. The maximum spectral radius across all nodes is 0.957, inside the unit circle, with a median of 0.081. No node produces an unstable fixed point. Together these results establish that the equilibrium characterised throughout the analysis is unique: the findings do not depend on equilibrium selection and there is no multiplicity problem to resolve.

{Equilbrium Uniqueness Table]

Pressure Scaling Robustness

The SMM recovers α_p at its lower bound, raising a question about whether the pressure irrelevance finding reflects genuine strategic structure or a cardinal scaling artefact. If pressure is always small relative to costs and benefits by construction, the GSA varying pressure multiplicatively around a near-zero baseline may never give it sufficient weight to matter regardless of how the strategic environment shifts. To test this directly, α_p is forced to match α_c in cardinal scale, giving pressure equal absolute weight to transition costs. If pressure correlations remain near zero even at this equalised scaling, the irrelevance finding is structural. If they become significant, the baseline result was driven by the low cardinal weight recovered by SMM rather than by the strategic mechanics of the model.

[Pressure equalised to cost vs Base comparison table]

[FIGURE: pressure_robustness]

The equalisation exercise confirms the pressure irrelevance finding is partially but not entirely a cardinal scaling artefact. Forcing alpha_p = alpha_c αp=αc activates the pressure channel: correlations that were statistically indistinguishable from zero at baseline become significant across multiple outcome dimensions, with pressure_US pressure_US reaching ρ=0.226 for coordination success and ρ=−0.132 for timing. The baseline result therefore cannot be attributed to strategic structure alone.

The policy hierarchy nonetheless survives the stress test. Costs and patience remain the dominant channels under equalisation, with mean absolute correlations of 0.184 and 0.135 respectively. Pressure rises from negligible to secondary, reaching a mean absolute correlation of 0.068, meaningful but well below the primary levers. The sweep evidence reinforces this ordering: pressure channels cross coordination frontiers under equalised scaling, but require reductions of 45 to 80% from an already inflated baseline value. In the fitted world, where alpha_p αp is small, those movements are not available.

The finding is therefore appropriately qualified. Diplomatic pressure and trade-linked conditionality are not strategically inert by construction. They are weak in the fitted 2025 environment because the cardinal weight the SMM assigns to the pressure channel is low relative to costs and benefits. Whether that weight is itself a feature of the current strategic environment or a limitation of the moment set is a question the present analysis cannot fully resolve. What it can establish is that even under the most favourable possible assumption about pressure's importance, it does not displace patience and cost reduction as the primary levers for compressing coordination failure risk.

Functional Form Robustness

Three functional form assumptions are stress-tested: the shape of collective benefits, the activation profile of political pressure, and the cost-learning curve. The central question in each case is whether the policy hierarchy identified in the main analysis survives the specification change.

The benefit specification is the only test where the hierarchy partially shifts. Linear and convex alternatives produce coordination success rates of 0.4% and 35.8% against the baseline 62.5%, and Spearman rank correlations of 0.393 and 0.334 against the baseline lever rankings. The quantitative ordering moves. The top of the hierarchy does not: theta, cost_US, and delta_US remain the dominant drivers under all three specifications. Patience and cost channels continue to dominate pressure and learning regardless of benefit form. The threshold finding is the one result that is genuinely specification-dependent, since its policy relevance requires benefits to concentrate near a participation frontier. For every other lever the hierarchy is robust.

The cost-learning curve produces no meaningful sensitivity. Across linear, convex, and concave alternatives, cost channels preserve their sign, magnitude, and significance throughout, with cost_US stable at approximately -0.33 across all three. The gamma correlation remains negligible regardless of specification. The cost hierarchy is fully invariant to how learning spillovers are modelled.

The pressure specification test is run jointly across functional form and cardinal scaling. At baseline alpha_p the hierarchy is unaffected: pressure correlations are flat across linear, convex, and sigmoid profiles and pressure plays no role in the lever ranking under any functional form. Under equalised alpha_p, pressure_US and pressure_RoW activate across all three specifications but remain well below costs and patience in magnitude. The policy hierarchy is therefore robust to pressure functional form at both cardinal scales. Whether pressure belongs in the hierarchy at all depends on its cardinal weight, not on how its activation is modelled.

[FIGURE: fig_functional_form_robustness] for benefit

[FIGURE: fig_pressure_spec_test] for pressure

[FIGURE: fig_cost_learning_spec_test] for cost

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